



Problem-Solution Fit

Date	18 July 2026
Team ID	TBD
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Students • Teachers • Educational Institutions • Online Learners	6. CUSTOMER LIMITATIONS • Limited internet access • Budget constraints • Low-end devices • Limited AI knowledge	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros • Online quizzes • Manual viva • LMS assessments Cons • Time-consuming • Limited feedback • Not voice-based • Less personalized
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty evaluating concept understanding	9. PROBLEM ROOT / CAUSE [RC] • No automated concept evaluation	7. BEHAVIOR + ITS INTENSITY [BE] • Frequently uses online learning platforms

- Manual assessment takes time
- Lack of personalized feedback
- Inconsistent evaluation
-

- Manual feedback is slow
- Lack of semantic analysis
- Difficulty measuring understanding accurately

- Regularly submits assignments
- Prefers instant feedback
- Uses voice communication for learning

3. TRIGGERS TO ACT [TR]

- Need quick concept evaluation
- Online learning growth
- Time-saving assessment
- AI-based learning support

4. EMOTIONS BEFORE / AFTER [EM] Before

- Confused
- Nervous
- Unsure of understanding

After

- Confident
- Motivated
- Improved learning

10. YOUR SOLUTION [SL]

- AI-powered Voice Based Concept Understanding Analyser
- Speech-to-text using Whisper
- Semantic similarity using Sentence Transformers
- AI feedback using Gemini
- Dashboard and PDF report generation

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- College LMS
- Streamlit Web App
- Mobile/Desktop
- Internet

OFFLINE

- Classroom
- Lab Sessions
- Viva Voce
- Self-study